

SHP ASSET MANAGER

HARDWARE BUILD GUIDE

Laser Eye V2 TX · Cable Trigger · Mini RX V2 · Full ESP32 RX V2 · Pi
TX · ESP-NOW Bridge

Assembly & wiring reference for all supported hardware

CONTENTS

01 Overview (#overview)

02 Laser Eye V2 TX (#laser-eye-tx)

03 Cable Trigger (#cable-trigger)

04 Mini RX V2 (#mini-rx)

05 Full ESP32 RX V2 (#full-rx)

06 Pi TX (#pi-tx)

07 ESP-NOW UART Bridge (#bridge)

08 Troubleshooting (#troubleshooting)

OVERVIEW

The SHP Asset Manager system is built around several complementary hardware units. You only need to build the devices that apply to your workflow – most users build one transmitter and one or two receivers.

LASER EYE V2 TX

V2

ESP32-based LiDAR transmitter. Detects subject distance and fires camera triggers over BLE or ESP-NOW. The primary triggering device for wildlife and nature photography.

CABLE TRIGGER

ESP32-based servo controller for physical cable release operation. Used when a wired shutter connection is needed. Pairs with the Cable Trigger tab in the app.

MINI RX V2

V2

Compact ESP32 receiver. Accepts ESP-NOW trigger packets from the Laser Eye V2 TX and drives a camera shutter output. Mounts close to the camera.

FULL ESP32 RX V2

V2

Full-size ESP32 dev-board receiver. Functionally identical to the Mini RX V2 – use this if you prefer a standard 38-pin form factor.

PI TX

PI

Raspberry Pi-based transmitter. Supports TF-Luna LiDAR, Pi Camera, USB webcam, DSLR via gPhoto2, Google Drive upload, time-lapse, astronomy, and live preview. Control it from the native PI TX app tab or the browser panel.

ESP-NOW UART BRIDGE

Attaches to the Pi TX's UART port and relays ESP-NOW trigger packets to Mini RX / Full RX receivers in the field – extends Pi TX range wirelessly.

WHAT EACH HARDWARE UNIT CAN DO

HARDWARE	CAPABILITIES	WORKS BY ITSELF?
Laser Eye V2 TX	Measures subject distance with TF-Luna LiDAR; fires its wired shutter and wake/AF outputs; drives the alignment laser; applies distance limits, smoothing, flux filtering, shot delay, cooldown, and shots-per-trigger; runs daily trigger windows; reports battery and sensor status; connects over BLE or its WiFi hotspot; pairs with up to four V2 receivers; can cross-link directly to a Cable Trigger; and supports OTA firmware updates.	Yes – it can trigger a camera through its wired output. Receivers and Cable Trigger links are optional.

HARDWARE	CAPABILITIES	WORKS BY ITSELF?
Cable Trigger	Physically presses a camera's cable release with its servo; controls delay, hold/exposure time, and engagement amount; stores presets; calibrates shutter timing with the phone microphone; accepts Android volume-button or iOS BLE-clicker remote firing; runs an autonomous longevity test; connects over BLE or WiFi; and can be triggered directly by a paired Laser Eye over ESP-NOW.	Yes – it can mechanically trigger a compatible cable release without a Laser Eye. The app is needed for setup and control.
Mini RX V2	Receives Laser Eye ESP-NOW trigger, wake/AF, and burst packets; drives an isolated camera shutter and wake/focus output; stores its pairing; reports battery, receiver type, and firmware; supports configurable wake polarity, AF pre-pulse, and LED behavior; and acknowledges received triggers.	No – it is a wireless camera-end receiver and needs a paired transmitter.
Full ESP32 RX V2	Provides the same wireless camera-end functions as the Mini RX V2 in a full-size ESP32 development-board format, with wired trigger/wake outputs, battery monitoring, pairing storage, configurable wake/AF behavior, status LEDs, and trigger acknowledgements.	No – it is a wireless camera-end receiver and needs a paired transmitter.
Pi TX 	Measures distance with TF-Luna LiDAR; drives shutter and wake/AF GPIO outputs (use the documented external isolation circuit for camera wiring); triggers DSLR/mirrorless cameras over USB with gPhoto2; captures from CSI Pi Cameras and UVC webcams; provides Pi Camera 3 focus, exposure, gain, and white-balance controls; serves a browser panel; runs time-lapses and astronomical schedules; stages and uploads photos to Google Drive; provides live preview; and can relay triggers to ESP-NOW receivers through the UART Bridge. Its secured <code>PI_TX_...</code> hotspot provides local control at <code>10.42.0.1</code> without internet.	Yes – LiDAR, camera, schedules, and local browser/phone control run locally. Internet is only needed for installation, Google Drive authorization/uploads, or updates.

HARDWARE	CAPABILITIES	WORKS BY ITSELF?
ESP-NOW UART Bridge	Connects to a Pi TX over UART; pairs with and manages Mini RX / Full RX receivers; relays trigger, wake, burst, battery, receiver-info, and configuration packets; keeps receiver pairing in its own memory; and supports secured OTA firmware updates through the Pi TX and app.	No — it is a Pi TX expansion and needs a Pi TX plus receiver hardware.

The SHP Asset Manager mobile app is the control surface for the ESP32 units and Pi TX. It is software rather than a hardware unit, so its controls are described in the user manuals.

This guide covers assembly and wiring only. For firmware flashing and software installation see the [Firmware & Software Install Guide](#). For app usage see the [Lazer Eye User Manual](#).

LASER EYE V2 TX

v2

The Laser Eye V2 TX runs on any standard ESP32 dev board (38-pin NodeMCU-32S or equivalent). It reads distance from a TF-Luna LiDAR sensor over I²C and fires a trigger output when a subject enters the configured range.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
ESP32 dev board	38-pin, NodeMCU-32S or equivalent. Must expose GPIO 2, 4, 5, 12, 18, 21, 22, 34.	1
TF-Luna LiDAR	I ² C interface. Operates on 3.3 V or 5 V.	1
LiPo battery	3.7 V single cell. Capacity to suit your session length.	1
LiPo charge/protect board	TP4056 or equivalent with protection circuit.	1
Voltage divider resistors	R1 = 47 kΩ, R2 = 100 kΩ (for battery ADC).	2
Power LED	Any standard LED. Colour of your choice.	1
Power LED resistor	330 Ω.	1
Laser module (optional)	5 mW dot laser for alignment. Any 3.3 V-compatible module.	1
2N3904 NPN transistor or PC817 optocoupler	For isolated trigger output. See wiring notes below.	1
1 kΩ base resistor	For transistor-based trigger isolation.	1
Camera shutter cable	2.5 mm TRS or 3.5 mm TRRS to suit your camera.	1
Enclosure / case	3D-printed or off-the-shelf weatherproof box.	1

GPIO PINOUT

GPIO	SIGNAL	CONNECTS TO
GPIO 2	STATUS_LED	Onboard LED (active HIGH). No external wiring needed.
GPIO 4	TRIGGER_OUT	Isolation circuit → camera shutter (tip/ring).

GPIO	SIGNAL	CONNECTS TO
GPIO 5	LASER	Laser module anode (via 33 Ω if module lacks its own resistor).
GPIO 12	WAKE_OUT	Isolation circuit $\rightarrow$ camera wake/AF (ring). GPIO 16 fires identically as an alternate.
GPIO 18	POWER_LED	LED anode $\rightarrow$ 330 Ω $\rightarrow$ GPIO 18. Cathode $\rightarrow$ GND.
GPIO 21	SDA	TF-Luna SDA pin.
GPIO 22	SCL	TF-Luna SCL pin.
GPIO 34	BATT_ADC	Battery voltage divider midpoint. Input-only; no pull-up.

TF-LUNA LIDAR WIRING (I²C)

TF-LUNA PIN	ESP32 PIN	NOTES
VCC	3.3 V or 5 V	TF-Luna tolerates either; use 5 V for full rated range.
GND	GND	
SDA	GPIO 21	ESP32 internal pull-ups are sufficient.
SCL	GPIO 22	

The TF-Luna must be in I²C mode. If your unit shipped in UART mode, short its UART/I²C mode pad or use the TF-Luna config tool to switch it before assembly.

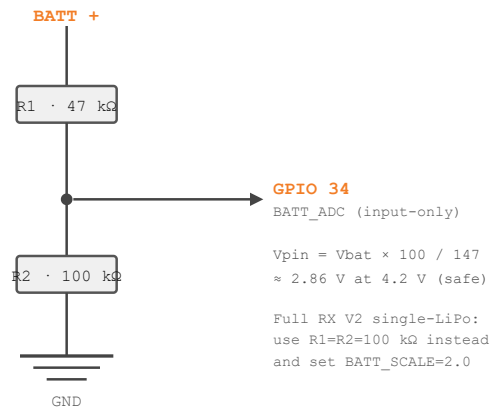
BATTERY ADC VOLTAGE DIVIDER

Wire a resistor divider between battery positive and GND. Tap the midpoint into GPIO 34.

NODE	CONNECTION
Battery +	R1 top (R1 = 47 k Ω)
R1 bottom / R2 top	GPIO 34
R2 bottom (R2 = 100 k Ω)	GND

This divider gives $V_{pin} = V_{bat} \times 100 / 147$. At a full 4.2 V cell that is ~2.86 V – safely inside the ESP32's 3.3 V ADC range.

BATTERY VOLTAGE DIVIDER – LASER EYE TX / FULL RX V2



TRIGGER OUTPUT – ISOLATION CIRCUIT

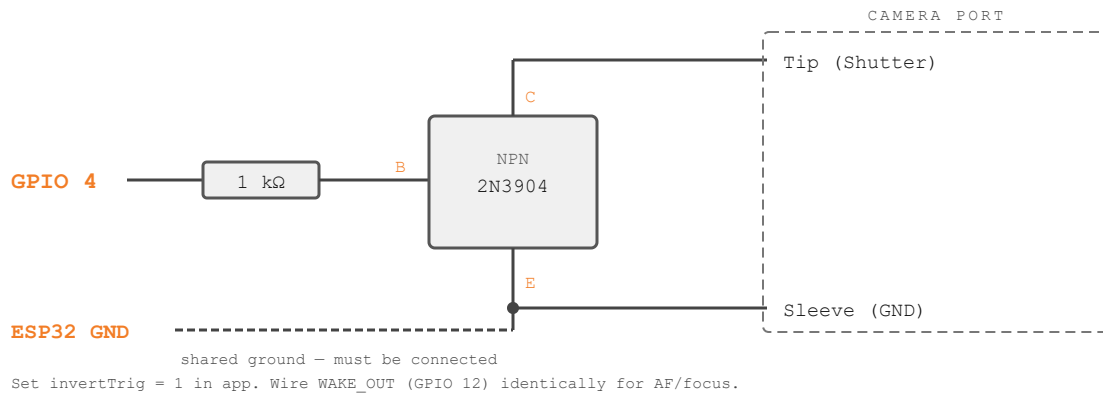
Never connect GPIO directly to your camera's shutter port without isolation. A wiring mistake could damage the camera. Use Option A only for quick bench testing.

Option A – Direct (bench testing only): GPIO 4 → camera shutter tip. ESP32 GND → camera sleeve (GND). GPIO 4 pulls LOW to trigger, short-circuiting the tip to GND. Set `invertTrig = 0` in firmware.

Option B – NPN transistor (recommended):

- > GPIO 4 → 1 kΩ → Base of 2N3904
- > Collector → camera shutter tip
- > Emitter → camera shutter GND (sleeve)
- > Camera GND and ESP32 GND must share a common reference
- > Set `invertTrig = 1` in the app (trigger fires LOW)

OPTION B – NPN TRANSISTOR ISOLATION CIRCUIT (GPIO 4 → CAMERA)



Option C – PC817 optocoupler (full isolation):

- > GPIO 4 → 330 Ω → PC817 anode (pin 1)
- > PC817 cathode (pin 2) → GND
- > PC817 collector (pin 4) → camera shutter tip
- > PC817 emitter (pin 3) → camera shutter GND
- > Camera side is fully isolated – no shared ground required
- > Set `invertTrig = 1` in the app

Wire the WAKE_OUT (GPIO 12 or GPIO 16) to the camera's AF / focus ring using the same circuit.

CABLE TRIGGER

The Cable Trigger drives an FS90MR continuous-rotation micro servo that physically actuates a cable release. It includes a TM1637 LED display and up to six tactile buttons for manual operation.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
ESP32 dev board	Any standard 30- or 38-pin ESP32. Must expose GPIO 18, 19, 22, 23, 25, 26, 27, 32, 33.	1
FS90MR servo	Continuous-rotation micro servo. Signal on GPIO 18.	1
5 V power supply	Capable of at least 500 mA for the servo. USB power bank or regulated 5 V.	1
TM1637 4-digit display	Common-cathode, 0.36" or 0.56" digit. CLK = GPIO 27, DIO = GPIO 32. Required – the firmware unconditionally drives it.	1
Tactile push buttons	Standard 6×6 mm or 12×12 mm. Up to 6 buttons.	1–6
10 kΩ pull-up resistors	One per button if not using internal pull-ups.	1–6
Cable release	Mechanical cable release with locking ring, to suit your lens or camera body.	1
Enclosure	3D-printed or off-the-shelf box.	1

GPIO PINOUT

GPIO	SIGNAL	CONNECTS TO
GPIO 2	STATUS_LED	Onboard LED (active HIGH). Optionally wire an external LED: anode → 330 Ω → GPIO 2, cathode → GND. See LED behaviour below.
GPIO 18	SERVO_SIGNAL	FS90MR signal wire (usually orange or white).
GPIO 19	BUTTON_1	Tactile button → GND. Internal pull-up enabled in firmware.

GPIO	SIGNAL	CONNECTS TO
GPIO 22	BUTTON_3	Tactile button → GND.
GPIO 23	BUTTON_4	Tactile button → GND.
GPIO 25	BUTTON_5	Tactile button → GND.
GPIO 26	BUTTON_6	Tactile button → GND.
GPIO 27	TM1637_CLK	TM1637 display CLK pin.
GPIO 32	TM1637_DIO	TM1637 display DIO pin.
GPIO 33	BUTTON_2	Tactile button → GND.

STATUS LED BEHAVIOUR

LED PATTERN	MEANING
Two quick flashes at boot	Firmware running – confirms the board powered up correctly.
Slow heartbeat (100 ms on / 1.9 s off)	Advertising over BLE – waiting for the app to connect.
Steady on	BLE connected and idle – ready to trigger.
Fast blink (125 ms on / 125 ms off)	Trigger active – servo is moving or holding.
Off	Deep sleep.

GPIO 2 drives the onboard LED directly. To add an external indicator LED, wire it anode → 330 Ω resistor → GPIO 2, cathode → GND. Both the onboard and external LED will light simultaneously.

SERVO POWER WIRING

The FS90MR draws up to 300 mA under load. Do not power it from the ESP32's 3.3 V regulator. Use a separate 5 V supply or the raw USB/VIN rail.

SERVO WIRE	CONNECTS TO
Signal (orange/white)	GPIO 18
Power (red)	External 5 V supply positive
GND (brown/black)	External 5 V supply GND and ESP32 GND (shared ground required)

TM1637 DISPLAY

The TM1637 display is a required hardware component. The firmware unconditionally initialises and writes to it – the sketch will not compile without the TM1637 library, and the display must be wired for correct operation.

DISPLAY PIN	ESP32 PIN
CLK	GPIO 27
DIO	GPIO 32
VCC	3.3 V or 5 V (module dependent)
GND	GND

BUTTON WIRING

Each button wires between its assigned GPIO and GND. The firmware enables the ESP32's internal pull-up resistors, so no external resistors are required.

Unused button GPIOs do not need to be connected to anything – they are ignored by the firmware when no button is detected.

MINI RX V2

v2

The Mini RX V2 runs on a compact ESP32 module and listens for ESP-NOW trigger packets from the Laser Eye V2 TX. It drives camera shutter and wake outputs through a transistor or optocoupler isolation circuit.

The firmware supports both original ESP32 DevKit Mini (non-S2) and ESP32-S2 Mini boards. The S2 variant uses a different battery ADC pin and exposes alternate output pins on the odd-column header for a cleaner single-connector build.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
ESP32 Mini dev board	Original ESP32 DevKit Mini or ESP32-S2 Mini. See pin differences below.	1
LiPo battery	3.7 V single cell.	1
LiPo charge board	TP4056 or equivalent with protection.	1
2N3904 or 2N2222 NPN transistor (×2)	One each for TRIGGER_OUT and WAKE_OUT.	2
1 kΩ base resistors	One per transistor.	2
Power LED + 330 Ω resistor		1 each
Camera shutter cable	2.5 mm TRS or 3.5 mm TRRS to suit your camera.	1
Battery voltage divider (non-S2 only)	R1 = 47 kΩ, R2 = 100 kΩ.	2
2N3904 switched divider (S2 only)	R1 = 47 kΩ (collector load), R2 = 100 kΩ (emitter to GPIO 9), base resistor = 10 kΩ to GPIO 12.	—

GPIO PINOUT — ORIGINAL ESP32 MINI (PRIMARY PINS)

GPIO	SIGNAL	CONNECTS TO
GPIO0	PAIR_CLR_PIN	Boot button (already on board). Hold LOW for 3 s on power-on to clear pairing.

GPIO	SIGNAL	CONNECTS TO
<code>GPIO 2</code>	<code>STATUS_LED</code>	Onboard LED (active HIGH). No external wiring needed.
<code>GPIO 4</code>	<code>TRIGGER_OUT</code>	Transistor/opto isolation circuit → camera shutter tip.
<code>GPIO 5</code>	<code>WAKE_OUT</code>	Transistor/opto isolation circuit → camera wake/AF ring.
<code>GPIO 18</code>	<code>POWER_LED</code>	LED anode → 330 Ω → GPIO 18. Cathode → GND.
<code>GPIO 34</code>	<code>BATT_ADC</code> (non-S2)	Midpoint of R1/R2 voltage divider. Input-only.

GPIO PINOUT — ESP32-S2 MINI (ALT CONNECTOR PINS)

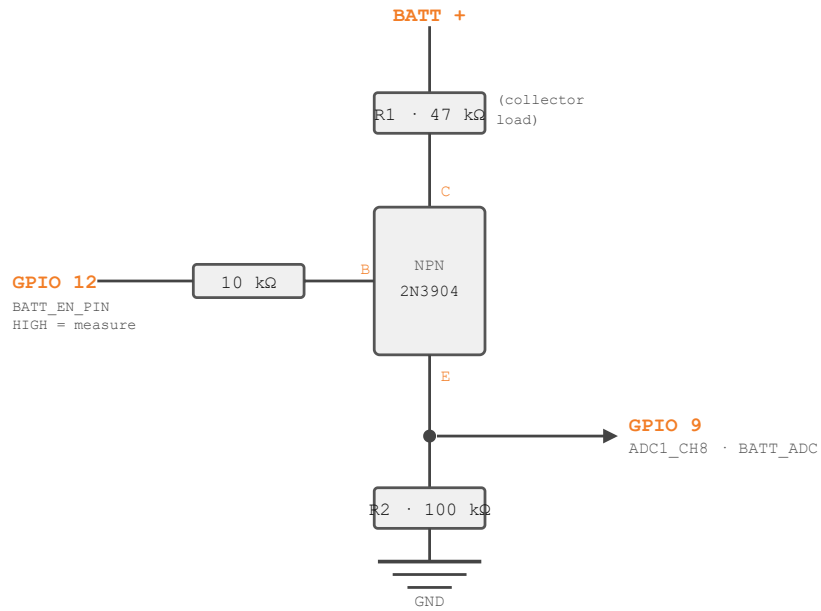
The S2 Mini's odd-column header exposes GPIOs 3, 5, 7, and 11 in a single row. Using these alternate pins lets you route all camera signals through one connector.

GPIO	SIGNAL	NOTES
<code>GPIO 3</code>	<code>STATUS_LED_ALT</code>	Fires simultaneously with GPIO 2. Use either or both.
<code>GPIO 7</code>	<code>TRIGGER_OUT_ALT</code>	Fires simultaneously with GPIO 4. Use either or both.
<code>GPIO 11</code>	<code>POWER_LED_ALT</code>	Fires simultaneously with GPIO 18. Use either or both.
<code>GPIO 9</code>	<code>BATT_ADC</code> (S2 only)	ADC1_CH8. Emitter of 2N3904 switched divider. Use instead of GPIO 34 on S2.
<code>GPIO 12</code>	<code>BATT_EN_PIN</code> (S2 only)	2N3904 base drive via 10 k Ω . HIGH enables the divider for a reading.

BATTERY ADC — S2 PIN ASSIGNMENTS

The S2 Mini lacks ADC on GPIO 34. The firmware routes the battery sense signal through GPIO 9 (ADC1_CH8) and uses GPIO 12 to gate the measurement circuit on only when a reading is needed. The exact resistor-and-transistor circuit is documented in the firmware header comment at the top of `firmware/lr-mini-rx-v2/lr_mini_rx_v2_firmware.ino` — build the circuit exactly as shown there, including component values and topology.

S2 MINI SWITCHED ADC - BATTERY VOLTAGE SENSE CIRCUIT



TRIGGER AND WAKE OUTPUT WIRING

Use a transistor isolation circuit for each camera output (same as Laser Eye V2 TX – see [Section 02](#)). GPIO 4 drives the shutter (TRIGGER_OUT) and GPIO 5 drives wake/AF (WAKE_OUT).

The Mini RX V2 firmware identifies itself to the Laser Eye V2 app as "MINI RX V2". On first pairing, it sends its device type back over ESP-NOW so the app can display it correctly.

FULL ESP32 RX V2

V2

The Full ESP32 RX V2 uses a standard 38-pin or 30-pin ESP32 dev board. It is functionally identical to the Mini RX V2 and uses the same pinout for all camera-facing signals. The main difference is form factor and the battery ADC circuit – full-size boards expose GPIO 34 as a true ADC input, so no switched divider is needed.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
ESP32 dev board	38-pin or 30-pin full-size. NodeMCU-32S, DOIT DevKit, or equivalent.	1
LiPo battery or 2×18650 parallel	See battery ADC note below for divider ratio.	1
Voltage divider resistors	R1 = 47 kΩ, R2 = 100 kΩ for 2×18650 parallel (default, BATT_SCALE 1.47). Use 100 kΩ / 100 kΩ for single LiPo (change BATT_SCALE to 2.0 in firmware).	2
2N3904 or 2N2222 NPN transistors (×2)	Shutter and wake isolation.	2
1 kΩ base resistors (×2)		2
Power LED + 330 Ω resistor		1 each
Camera shutter cable		1

GPIO PINOUT

GPIO	SIGNAL	CONNECTS TO
GPIO 0	PAIR_CLR_PIN	Boot button. Hold LOW 3 s on power-on to clear pairing.
GPIO 2	STATUS_LED	Onboard LED (active HIGH).
GPIO 4	TRIGGER_OUT	Isolation circuit → camera shutter tip.
GPIO 5	WAKE_OUT	Isolation circuit → camera wake/AF ring.
GPIO 18	POWER_LED	LED anode → 330 Ω → GPIO 18. Cathode → GND.

GPIO	SIGNAL	CONNECTS TO
GPIO 34	BATT_ADC	Voltage divider midpoint. Input-only ADC pin.

BATTERY ADC NOTES

The firmware ships with `BATT_SCALE 1.47`, calibrated for the 47 kΩ / 100 kΩ divider used with a 2×18650 parallel pack. For a single LiPo use equal 100 kΩ resistors and change `BATT_SCALE` to `2.0` in the firmware before flashing.

BATTERY TYPE	DIVIDER	BATT_SCALE	MAX VPIN AT 4.2 V
2×18650 parallel (3.0-4.2 V)	R1 = 47 kΩ, R2 = 100 kΩ (default)	1.47 (default)	~2.86 V (safe)
Single LiPo (3.0-4.2 V)	R1 = 100 kΩ, R2 = 100 kΩ	2.0 (must update)	~2.10 V (safe)

If you use the single-LiPo equal-resistor divider without updating `BATT_SCALE` to 2.0, battery percentage readings will be wrong. Edit the define near the top of `lr_rx_v2_firmware.ino` before flashing.

The Pi TX turns a Raspberry Pi into a full-featured camera trigger and upload station. It supports TF-Luna LiDAR triggering, Pi Camera CSI, USB webcam, DSLR via gPhoto2, live preview, time-lapse, astronomical scheduling, and automatic upload to Google Drive. All control happens through the phone app or browser panel – no monitor or keyboard is needed in the field.

RECOMMENDED RASPBERRY PI MODELS

- > Raspberry Pi 4 Model B – best performance; handles simultaneous LiDAR + camera + upload.
- > Raspberry Pi 3 Model B+ – supported for standard trigger and camera workflows.
- > Raspberry Pi 5 – supported and best for heavier simultaneous camera and upload workloads.

The installer is tested on Raspberry Pi OS Lite (Bookworm, 64-bit) with Pi 3B+, Pi 4, and Pi 5. Other models may work but are not part of the current tested target.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
Raspberry Pi	Pi 3B+, Pi 4 Model B, or Pi 5 (see above).	1
microSD card	32 GB minimum; Class 10 / A1 or better.	1
5 V power supply	Official Pi PSU or a high-quality USB-C / micro-USB supply. 3 A recommended for Pi 4.	1
TF-Luna LiDAR (optional)	I ² C interface. Required only if using distance triggering.	1
Pi Camera Module (optional)	Pi Camera v2 or Pi Camera 3. Connected via CSI ribbon cable.	1
2N3904 NPN transistor or PC817 optocoupler	For isolated shutter/wake output (Option B/C). Recommended over direct connection.	1- 2
1 kΩ resistors	For transistor base drive.	1- 2
Camera shutter cable	2.5 mm TRS or 3.5 mm TRRS to suit camera.	1

COMPONENT	SPEC / NOTES	QTY
ESP-NOW UART Bridge (optional)	Separate ESP32 board. See Section 07 .	1

40-PIN GPIO WIRING

All references use physical pin numbers (as printed on breakout boards). GPIO numbers are shown in parentheses.

PHYSICAL PIN	GPIO	SIGNAL	CONNECTS TO
Pin 4	5 V	TF-Luna VCC	TF-Luna VCC (if using LiDAR)
Pin 6	GND	TF-Luna GND	TF-Luna GND
Pin 3	GPIO 2 (SDA)	I ² C SDA	TF-Luna SDA
Pin 5	GPIO 3 (SCL)	I ² C SCL	TF-Luna SCL
Pin 11	GPIO 17	SHUTTER	Isolation circuit → camera shutter
Pin 13	GPIO 27	WAKE / AF	Isolation circuit → camera wake/AF
Pin 8	GPIO 14 (TXD)	UART TX → Bridge RX	ESP-NOW Bridge GPIO 16 (optional)
Pin 10	GPIO 15 (RXD)	UART RX ← Bridge TX	ESP-NOW Bridge GPIO 17 (optional)
Pin 1	3.3 V	Bridge 3.3 V (optional)	ESP32 bridge 3.3 V (bare module only)
Pin 14	GND	Shared GND	Bridge GND; isolation circuit GND

SHUTTER OUTPUT — TRIGGER WIRING OPTIONS

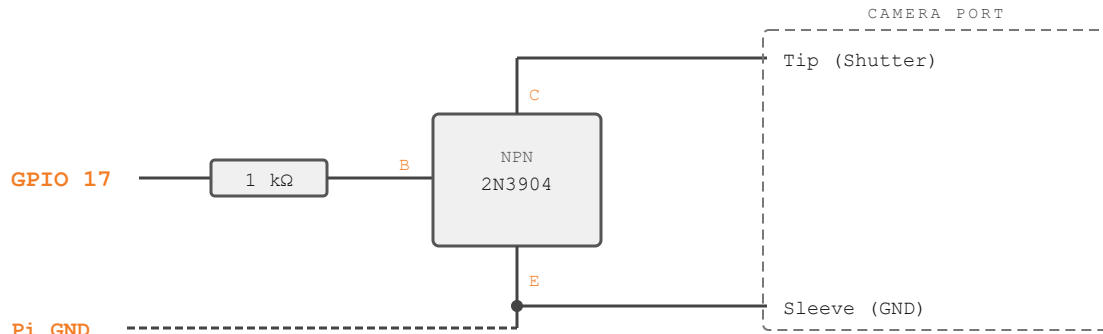
Do not connect the Pi's 3.3 V GPIO directly to a camera shutter without isolation. The Pi GPIO cannot source or sink significant current safely, and a wiring error can damage the Pi.

Option A — Direct dry-contact (bench testing only): GPIO 17 → camera shutter tip. Pi GND → camera sleeve (GND). GPIO 17 pulls LOW to fire, short-circuiting tip to GND and closing the shutter contact. No transistor. Set `invertTrig = 0` (default).

Option B — NPN transistor (recommended):

- > GPIO 17 → 1 kΩ → 2N3904 Base
- > Collector → camera shutter tip
- > Emitter → camera GND (sleeve)
- > Pi GND and camera GND must share a common reference
- > Set `invertTrig = 1` in the app

OPTION B – NPN TRANSISTOR ISOLATION CIRCUIT (GPIO 17 → CAMERA)



shared ground – must be connected
Set `invertTrig = 1` in app. Wire GPIO 27 (WAKE/AF) identically using the same circuit.

Option C – PC817 optocoupler (full isolation):

- > GPIO 17 → 330 Ω → PC817 anode (pin 1)
- > PC817 cathode (pin 2) → Pi GND
- > PC817 collector (pin 4) → camera shutter tip
- > PC817 emitter (pin 3) → camera shutter GND (sleeve)
- > Set `invertTrig = 1` in the app

Hot-mode trigger (trigMode = 1): GPIO 17 drives 3.3 V directly to the camera tip. Use only for cameras that accept an active-HIGH signal on the shutter port – verify your camera's spec before wiring.

Wire GPIO 27 (WAKE/AF) using the same isolation circuit as the shutter output.

RASPI-CONFIG SETTINGS

The install script runs `raspi-config` non-interactively for the following two settings. If you need to verify them manually:

- > **Interface Options** → **I²C**: Enable (required for TF-Luna LiDAR).

> **Interface Options** → **Serial Port** → **Login shell over serial**: Disable. **Hardware serial port**: Enable. (Required for UART Bridge.)

Pi Camera support does not require a separate raspi-config camera step on modern Raspberry Pi OS (Bookworm). The install script installs `python3-picamera2` and `libcamera-apps` via apt, which is sufficient for both Pi Camera v2 and Pi Camera 3 CSI modules.

LOCAL-FIRST WIFI NETWORK

The installer creates a secured NetworkManager hotspot that stays available for local control, even when the Pi has no internet. The SSID begins with `PI_TX_`, the Pi uses `10.42.0.1`, and the generated password is printed by the installer and stored root-only on the Pi. Ethernet or a second WiFi adapter can optionally provide upstream internet without taking the built-in WiFi adapter away from the hotspot.

Do not connect the Pi's GPIO directly to a camera shutter without the documented isolation circuit. The direct dry-contact option is for bench testing only.

ESP-NOW UART BRIDGE

The ESP-NOW UART Bridge connects to the Pi TX's hardware UART port and re-transmits trigger packets over ESP-NOW to any paired Mini RX V2 or Full RX V2 receivers in the field. It lets the Pi TX drive wireless receivers without any dedicated ESP-NOW radio on the Pi itself.

SUPPORTED ESP32 BOARDS

- > **ESP32-DevKit-C / WROOM-32** – plug-and-play, no pin remapping needed.
- > **ESP32-C3 Mini** – change `BRIDGE_UART_RX` to GPIO 6 and `BRIDGE_UART_TX` to GPIO 7 in the firmware before flashing.
- > **ESP32-S3 Mini** – change `BRIDGE_UART_RX` to GPIO 4 and `BRIDGE_UART_TX` to GPIO 5 before flashing.

PARTS LIST

COMPONENT	SPEC / NOTES	QTY
ESP32-DevKit-C, WROOM-32, C3 Mini, or S3 Mini	See pin remap note above for mini variants.	1
Logic-level wires	3 wires: RX, TX, GND between Pi and ESP32.	—
3.3 V supply wire (bare module only)	If using a bare ESP32 module without an onboard 3.3 V regulator, power from Pi Pin 1.	—

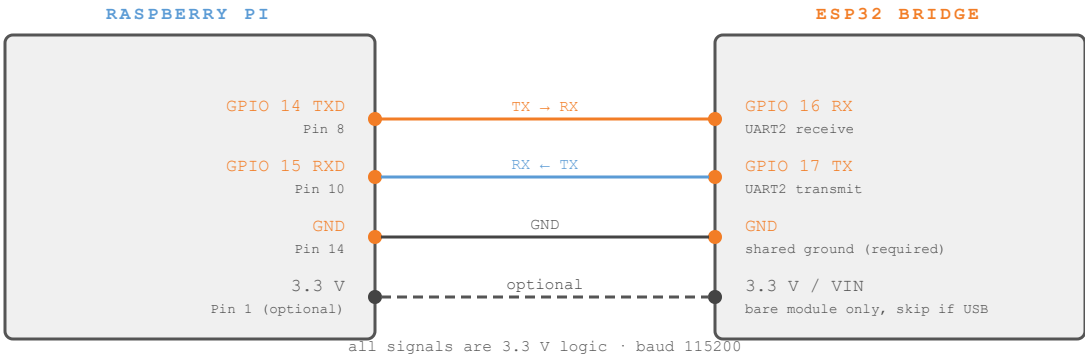
UART WIRING (DEVKIT-C / WROOM-32)

RASPBERRY PI	PHYSICAL PIN	ESP32 GPIO	SIGNAL
GPIO 14 (TXD)	Pin 8	GPIO 16 (UART2 RX)	Pi transmits → Bridge receives
GPIO 15 (RXD)	Pin 10	GPIO 17 (UART2 TX)	Bridge transmits → Pi receives
GND	Pin 14	GND	Shared ground (required)
3.3 V (optional)	Pin 1	3.3 V or VIN	Power (bare module only; skip if using USB)

The Pi's UART is 3.3 V logic. Do not connect it to a 5 V UART port – this will damage the Pi's GPIO. All standard ESP32 dev boards are 3.3 V logic compatible.

UART baud rate between Pi and Bridge is fixed at 115200 in both the Pi TX daemon and the bridge firmware. Both sides must match.

UART WIRING – RASPBERRY PI TX ↔ ESP32 BRIDGE (DEVKIT-C / WROOM-32)



PIN REMAP FOR C3 / S3 MINI VARIANTS

Before flashing the bridge firmware to a C3 or S3 Mini board, edit these two `#define` lines near the top of `pi_espnow_bridge.ino`:

BOARD	BRIDGE_UART_RX	BRIDGE_UART_TX
ESP32-DevKit-C / WROOM-32	GPIO 16 (default)	GPIO 17 (default)
ESP32-C3 Mini	GPIO 6	GPIO 7
ESP32-S3 Mini	GPIO 4	GPIO 5

The Pi TX UART wiring does not change – only the ESP32 pin assignments in firmware need updating for mini variants.

TROUBLESHOOTING

Common hardware assembly problems and their fixes. Firmware-related issues are covered in the [Install Guide §12](#).

ESP-NOW PAIRING

Laser Eye V2 TX fires but the Mini RX V2 / Full RX V2 does not respond – no trigger output on the receiver

Confirm the receiver's STA MAC address was entered correctly during pairing in the app (format `AA:BB:CC:DD:EE:FF` – all uppercase, colon-separated). The MAC is printed to Serial Monitor at 115200 baud on every boot. If the MAC was entered correctly, clear the pairing on both sides: hold the receiver's BOOT button (GPIO 0) LOW for 3 seconds on power-on until the LED blinks rapidly, then re-pair in the app. Ensure both devices are within reliable ESP-NOW radio range (open air: <200 m; walls and obstructions reduce this significantly).

Mini RX V2 shows as paired in the app but stops receiving triggers after a power cycle

The pairing is stored in flash (`Preferences`) on the receiver. If the receiver has been reflashed since pairing, the stored peer MAC is erased and you must re-pair. After any firmware flash, always re-pair the receiver from the app before use.

TRIGGER OUTPUT

Trigger output not firing the camera – shutter does not activate even though the trigger LED blinks

Check the `invertTrig` setting in the app – the correct value depends on your wiring option. Direct connection (Option A): set `invertTrig = 0`. NPN transistor (Option B) or PC817 optocoupler (Option C): set `invertTrig = 1`. Also verify your camera's shutter port is 2.5 mm TRS (most DSLRs and mirrorless) and that tip and sleeve assignments match your wiring – a reversed cable will not fire the shutter.

Trigger fires the camera but the camera body powers off or behaves erratically

The camera GND (sleeve) and ESP32 / Pi GND are not sharing a common reference, causing a floating or unexpected voltage on the shutter port. Ensure the shared ground wire is connected as shown in the NPN transistor circuit diagram in §02 or §06. If using Option A (direct), ensure ESP32 GND is tied to the camera's shutter sleeve directly.

WAKE / AF output does not wake the camera before the shutter fires

Wire the WAKE_OUT pin (GPIO 12 on Laser Eye TX; GPIO 27 on Pi TX) to the camera's ring contact (2.5 mm TRS: ring = AF/wake; tip = shutter). Use the same isolation circuit as the shutter output. The firmware pulses WAKE_OUT before TRIGGER_OUT by the wake-up delay configured in the app – confirm this delay is non-zero.

BATTERY PERCENTAGE

Battery percentage reads 0 %, stuck at 100 %, or wildly inaccurate

Verify the voltage divider is wired correctly: Battery+ → R1 → midpoint → R2 → GND, with the midpoint tapped to the ADC input pin (GPIO 34 on full-size boards; GPIO 9 on ESP32-S2 Mini). Measure the midpoint voltage with a multimeter at a known battery charge level and confirm it matches the expected value from the divider formula. Also verify the resistor values match the values expected by the firmware's `BATT_SCALE` constant – a wrong resistor value (e.g. 10 kΩ instead of 100 kΩ) will produce consistently wrong readings.

ESP32-S2 Mini always reads 0 % battery regardless of charge level

The S2 Mini uses a switched ADC circuit controlled by GPIO 12 (BATT_EN_PIN). Confirm GPIO 12 is wired to the 10 kΩ base resistor as shown in the S2 Mini switched ADC diagram in §04. If GPIO 12 is unconnected, the transistor gate never opens and the ADC reads nothing. Also ensure you selected **ESP32-S2 Dev Module** as the board in Arduino IDE – compiling for a standard ESP32 board will use the wrong ADC pin assignments.

TF-LUNA LIDAR

TF-Luna not detected on I²C – Serial Monitor shows repeated "TF-Luna init failed" or distance always reads 0 or -1

First confirm the TF-Luna is in I²C mode – units shipped in UART mode will not respond on I²C. Use the TF-Luna configuration tool (TFMini Plus Config Tool, Windows only) or short the UART/I²C mode selection pad on the sensor PCB to switch it. Next check wiring: SDA → GPIO 21, SCL → GPIO 22 for Laser Eye TX; SDA → Pi GPIO 2 (Pin 3), SCL → Pi GPIO 3 (Pin 5) for Pi TX. The ESP32 uses internal pull-ups on I²C; the Pi may need external 4.7 kΩ pull-ups if the cable run is long (>30 cm). Finally confirm power: the TF-Luna draws up to 130 mA at 5 V – ensure the supply can deliver this alongside the rest of the circuit.

TF-Luna reports valid distance but readings are noisy or jump erratically

Strong ambient IR light (direct sunlight or nearby IR sources) saturates the TF-Luna's receiver and produces noisy readings. Shade the sensor aperture or reduce the

frame rate in firmware to give the sensor longer integration time. Also check that the I²C cable is away from the servo signal wire (Cable Trigger) or any PWM lines – coupling noise can corrupt readings.

[SHP Asset Manager](#) · [Hardware Build Guide](#)

[Firmware & Software Install](#) · [Lazer Eye Manual](#) · [Cable Trigger Manual](#)
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